MEDICAL AI PROJECT

Automated Chest X-Ray Disease Classifier

AI-powered system for detecting 14 chest diseases from medical X-ray images.

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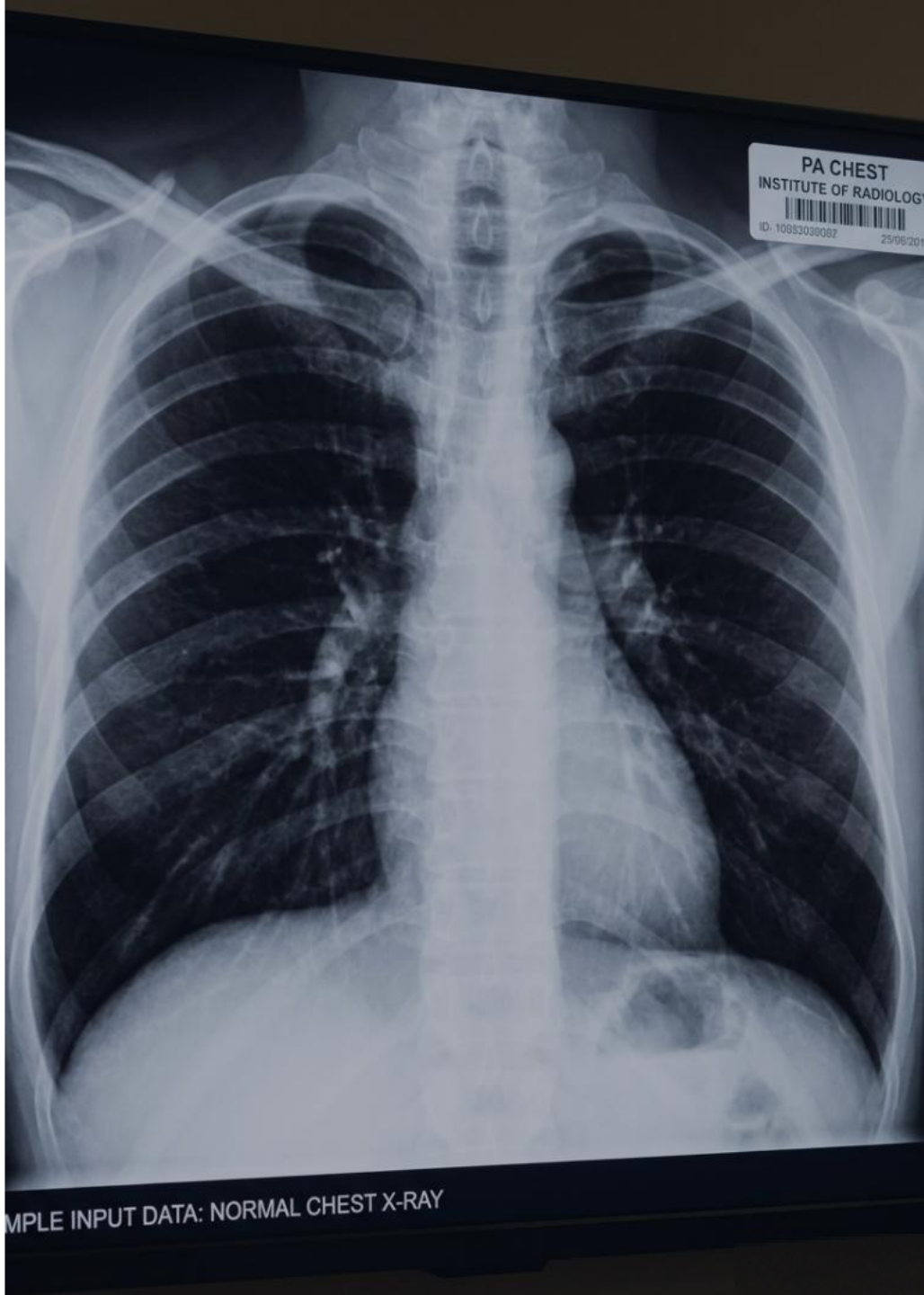
Machine Learning / AI Project

KEY TECHNOLOGIES

Python • PyTorch • Airflow • DVC



Problem Statement

The Challenge

- Diagnosing chest diseases from X-rays is highly time-intensive.
- Hospitals generate thousands of new imaging records every daily.
- Manual reading increases the risk of delays and human error.

Our Goal

Assist radiologists with AI that flags diseases automatically and accurately.

Project Overview

Powered by an extensive dataset of **112,120 medical X-ray images** for robust model training.

Process Images

Automated ingestion and preprocessing pipeline for raw clinical X-ray data.

Train AI Model

Trains a deep learning architecture to extract complex visual patterns.

Detect 14 Diseases

Outputs precise diagnostic probability scores across multiple pathologies.

Visualize Attention

Uses Grad-CAM heatmaps to show clinicians exactly where the AI is looking.

System Architecture

Dataset
Raw X-Ray Images

→

Data Pipeline
Processing & Augmentation

→

Deep Learning Model
Feature Extraction

→

Evaluation
Metrics & Predictions

EXECUTION STEPS

01 Data validation

02 Image preprocessing

03 Data augmentation

04 Train / Val / Test split

05 Model training & eval

Orchestrated by Apache Airflow
Automates the entire data preparation and model training workflow for complete reproducibility.

Dataset: NIH ChestX-ray14

112,120
Total X-Ray Images

30,805
Unique Patients

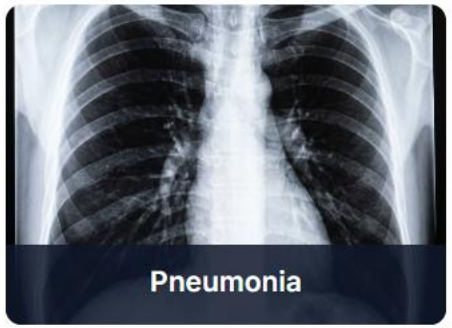
14 Disease Classes

Multi-label classification paradigm allowing for simultaneous detection of co-occurring conditions.

Key Pathologies Included

- Pneumonia
- Cardiomegaly
- Emphysema
- Edema
- Fibrosis
- Hernia

Sample X-Ray Scans



Data Engineering Pipeline

Automated via Apache Airflow

Purpose: Ensure absolute reproducibility and pipeline automation without manual intervention.

01

Validate Dataset

Verify image integrity and check label consistency.

02

Preprocess Metadata

Clean patient records and normalize feature formats.

03

Apply Augmentation

Rotate, scale, and adjust contrast to expand set.

04

Split Dataset

70% Train

15% Val / 15% Test

05

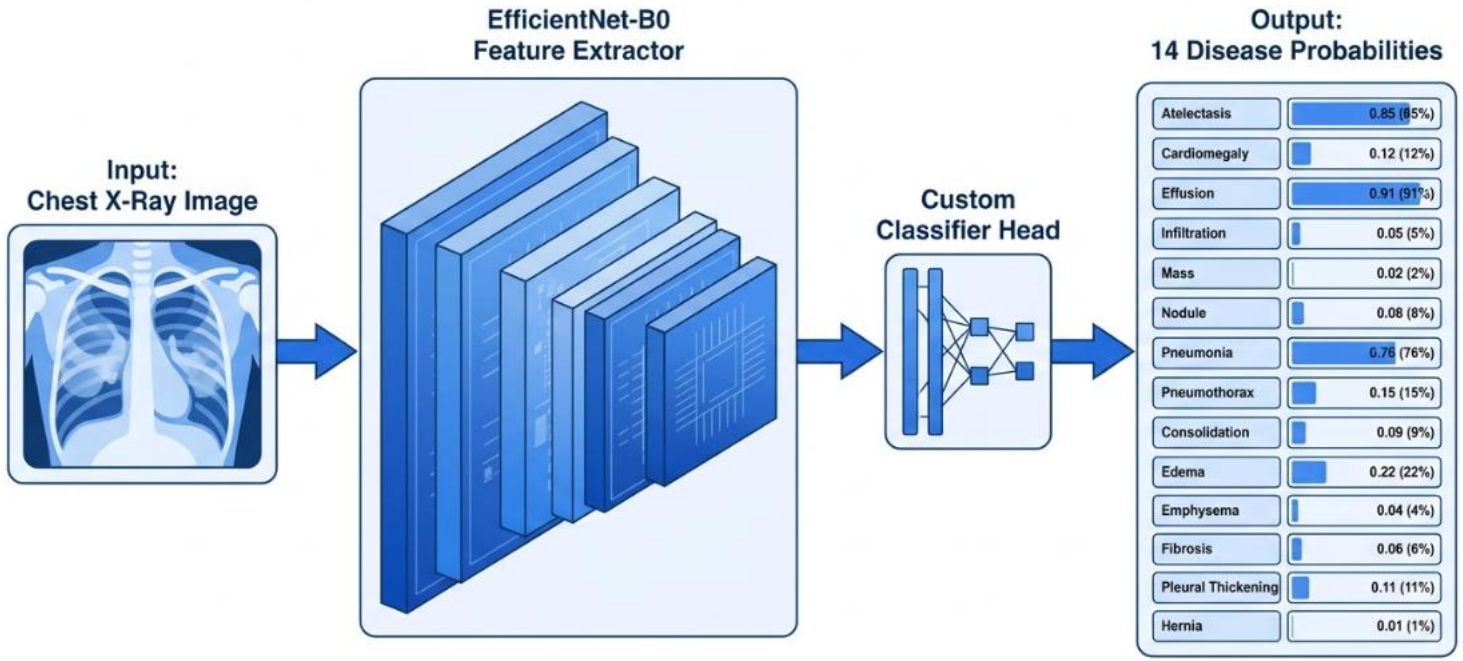
Version with DVC

Track data versions and model artifacts securely.

Model Architecture

A highly efficient convolutional neural network adapted for multi-label medical image classification.

ARCHITECTURE FLOW



EfficientNet-B0

Pretrained on ImageNet to extract complex visual patterns and features from chest X-rays.

Custom Classifier

Dense layer architecture predicting independent probabilities for 14 distinct diseases.

Training Configuration

Loss: Binary Cross Entropy

Optimizer: Adam

Hardware: NVIDIA T4 GPU

Model Performance

Robust diagnostic capabilities demonstrated across 14 independent chest pathologies.

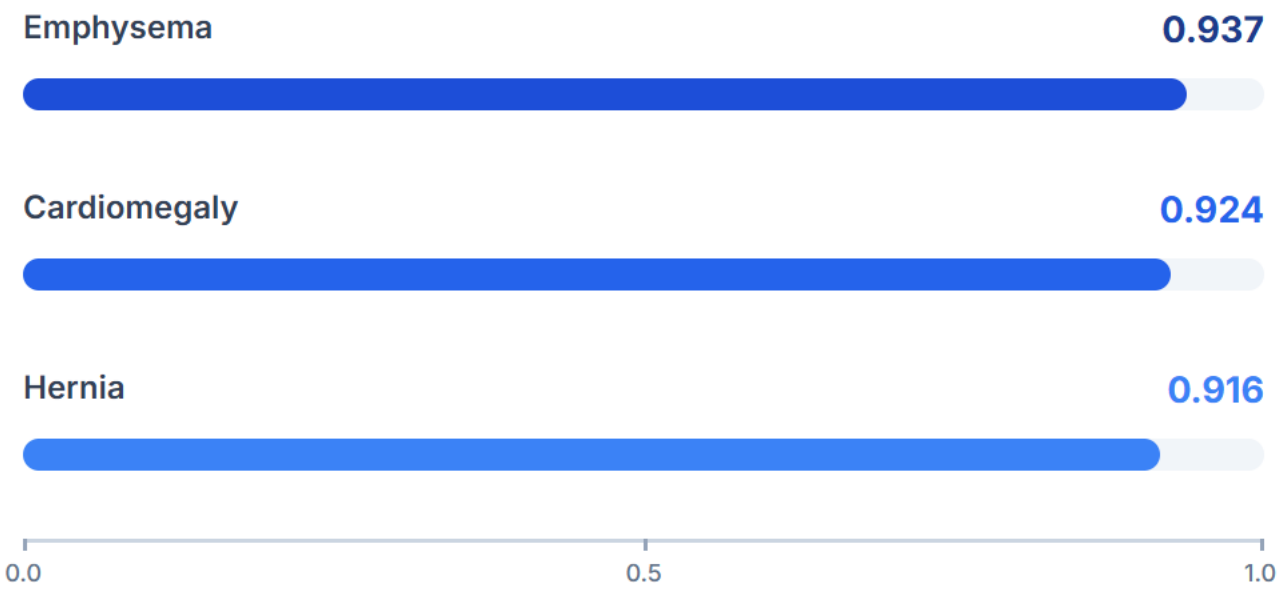
Overall Mean AUC-ROC

0.847

What is AUC-ROC?

Measures the model's ability to distinguish between **disease vs. no disease**. 1.0 represents a perfect prediction, while 0.5 is random guessing.

Top Performing Disease Classes



Model Explainability

Using Grad-CAM to visualize model attention, ensuring transparent and trustworthy AI diagnosis.

Visualizing Attention

Generates a precise heatmap overlay that highlights the exact pulmonary regions most influencing the AI's final prediction.

Building Clinical Trust

Transforms a "black box" neural network into an interpretable clinical tool, assuring radiologists of the model's logical reasoning.

Clinical Example: Emphysema Detection

In the adjacent image, the model correctly focuses its attention (red/orange areas) on the lower right lung field, aligning perfectly with genuine clinical pathology rather than image artifacts.

Model Attention

High (Focus)

Low (Ignore)

Technologies Used

A modern, production-ready stack powering the end-to-end machine learning pipeline.

Python
Core Language

PyTorch
Deep Learning

EfficientNet
Vision Backbone

Apache Airflow
Pipeline Orchestration

DVC
Data Versioning

OpenCV
Image Processing

Scikit-learn
Evaluation Metrics

Google Colab
Cloud Compute

Key Achievements

Delivering measurable impact through end-to-end automation and high-performance AI.

End-to-End ML Pipeline
Built a robust, production-style machine learning architecture from scratch.

Automated Data Orchestration
Engineered seamless data ingestion and preprocessing workflows via Apache Airflow.

Clinical-Grade Performance
Achieved over 85% Mean AUC-ROC, ensuring highly reliable multi-label diagnostics.

Transparent AI Diagnostics
Integrated Grad-CAM visualizations to provide clear model explainability for trust.

Reproducible Version Control
Deployed DVC for strict dataset versioning and perfectly reproducible model artifacts.

Project Impact

112,120
IMAGES PROCESSED

85.0%+
MEAN AUC-ROC

14
DISEASE CLASSES

Future Improvements

Roadmap for extending the clinical utility and deployment of the AI diagnostic system.

PHASE 1

Web Application

Deploy the model as an intuitive web interface for robust clinical testing, validation, and easy end-user access.

PHASE 2

Real-Time API

Add low-latency inference API endpoints to enable instant, on-demand automated evaluations of new chest X-rays.

PHASE 3

PACS/EHR Integration

Integrate smoothly with existing hospital workflows and electronic health record systems to deliver insights automatically.

PHASE 4

Expanded Datasets

Train the network on broader, multi-institutional datasets to enhance generalization across varied patient demographics.

AI empowering human expertise

By automating preliminary disease detection, this AI system can effectively assist radiologists to significantly improve diagnostic efficiency and accuracy.